Unit 23 - Human Computer Interaction
Assignment 3 - HCI Impacts

Introduction

Scenario / Vocational Context - *Your game company's initial releases have been hugely successful; as such your game development direction has become a high budget forward thinking process that spans many platforms and devices.*

You have become aware of the rise of mobile gaming and the approach of virtual and augmented reality interaction models. As such your company wants to complete some research into impacts contemporary and future HCI models may have on society, culture and the economy.

Task 1

(U23P1) - Describe the impact of HCI on society, the economy and culture

Task description - *Below is a list of contemporary interaction models:*

- *Touch-screen*
- *Virtual Reality*
- *Augmented Reality*
- *Voice*
- *Gesture*
- *Haptic*

Using examples of the above interaction models, ***describe*** *the negative and positive impacts on the following:*

- *Society (Work / Home Life, citizens (age/needs), conflict and nature)*
- *Economy (Gross Domestic Product, automation, employment, tax, spending & money)*
- *Culture (Ways of life, progressiveness, traditions and technology use/reliance.)*

A report that describes the impact on society, culture and the economy of the three interaction models provided

Introduction

The way we now interact with computers has undoubtedly revolutionised the world that everyone currently lives in. Due to the rapid development of Human-Computer Interaction (HCI), the modern world has become an increasingly efficient place with automated factory assembly lines, online banking, and self-service tills in supermarkets.

No longer are we required to complete unnecessarily difficult tasks, which could be done easier using some sort of device, making people's lives much easier at home or work. With modern culture and the economy also revolving around technology and HCI.

Yet, the more automated and efficient it becomes, the more potential problems we run into as a society, which could also cause different problems with our culture and even the economy.

Society (Work life, home life, citizens age, citizens needs, conflict & nature)

In modern society, we've without a doubt come a long way in terms of computer usability during work and home life. For instance, during the 1940s, only specialised computer scientists would have had access to a computer, and even throughout the 1960-80s, computers weren't very widespread.

Now, ever since the massive developments in technology and HCI, one of the main concerns has been how users interact and respond to computers. So, that is the reason modern technology is developed specifically to be beginner-friendly, and constantly improving it.

As computers become more prevalent and older generations start to pick up new technology, the need for improved user-friendliness has increased. User Interfaces on new touch screen phones, tablets and computers are designed with this in mind.

This has had an effect in today's society because it has made people more reliant on technology, debatable if there is such thing as being "too" reliant on technology. There are a number of positive benefits that have come about due to the developments in HCI. However, there are also a number of negative effects which have changed our society both in the present and in the future.

Recent updates like Siri for iPhone, Cortana for Windows phones and the more recent Amazon Echo product have opened up new possibilities for blind people using technology as they allow voice control for the basic features such as calling, sending texts/emails, and searching the Internet.

Since a large portion of the population now owns and uses some sort of computing device which can range from a smartphone, tablet, gaming console, laptop or a personal computer, sometimes a household can contain all these devices and possibly much more.

The effect of this modernisation and increased accessibility means that everybody is on a somewhat similar level when it comes to technology. Meaning that everybody has access to the same opportunities, beginners and experts alike.

Specialist forms of communications such as braille, text-to-speech, and voice commands have allowed blind people to have access to computers or help with everyday life when needed. For

instance, technology can be found in many public places such as hospitals, transportation, schools/colleges and most workplaces.

A good example of this are Amazon's Echo AI, which is widely available for the public which most of the population in first-world, maybe even developing, countries could afford, with voice recognition, so there is no need to access the hardware, and text-to-speech. Amazon is the only major company in the world to deploy this type of system, yet others are following suit, such as Google.

The effect of this is that blind people are then able to do the same things that people who can see can do within the household and even the workplace. Basically, it makes life a bit easier for those who are blind.

Other developments in HCI have been to tackle other, perhaps more demanding "disabilities", like autism for example. Even though it is a controversial topic whether autism is a disability, it does hinder certain learning, behaviour or social abilities to those that are found in the spectrum.

Hence why in recent years, scientists have tried developing VR software to improve social skills and brain activity to those with autism. Essentially, these are random scenarios inside a virtual world which requires the user to interact with the virtual people.

They are normally equipped with a webcam projecting his/her facial expressions while using the VR headset, with supervision of an autism therapist. This means that children with autism could improve on their social skills, making them feel less excluded from places like school where they can then learn more effectively.

Essentially, people that suffer from life-changing disabilities/conditions/ailments are less excluded from society because of these HCI developments which allow them to access the modern technology in an ever-changing, ever-modernising world. Tasks that would have been thought impossible but 20 years ago are now viable options for impaired people, and still improving.

<u>Economy (Gross Domestic Product, automation, employment, tax, spending & money)</u>

Throughout the years as the development with human computer interaction have introduced fresh and innovative devices to the market. Especially with the growth in society getting involved, making technology has grown significantly compared to the past at a consistent rate even to the present. It is without a doubt that developments in HCI have had a huge impact, not only on our culture and society, but on our economy too.

With the introduction of online banking, shopping and stock brokering, they have impacted the larger sections of the economy. For instance, the economy has become more manipulative in

recent years. With websites usually storing user data, which is then sold on to advertising companies to tailor to the user's specific needs. Where advertisers can target users with specific ads, thereby luring them in to purchase items that they might want or need.

This data is similar to the old fashioned way surveys questionnaires in some ways, since they allow the surveyor to collect useful business data. However, this time the user may not know that their data that the business managed to obtain, was actually obtained.

The benefit of predictability is that it encourages people to buy things from companies, which benefits the economy. However, this may cause some people to question the morality and this type of user manipulation is a form of corruption, especially without consent from the public.

In one hand, before the internet, the only method of purchasing groceries and other products was to go to a shop and physically buy it. Those days are not necessarily gone, but are without a doubt fading.

For example, companies that were once big such as HMV and GAME, have resorted to becoming online stores because keeping physical stores open. Which did cost a lot of money and the companies ended up going bankrupt due to the absurd budget.

This means that in modern day, the DIY method is to buy products online. Meaning, with a few clicks of a button, the user can purchase their weekly groceries and other luxuries or essentials.

Even with some stores including a way of saving the user's recently purchased items or creating a list of desired items. This list can then be used as data, which would manipulate the items seen by the user, going back to the same things which the user normally purchases. A few of the biggest companies that currently do this are Tesco and Ocado, they also were one of the first.

However, this could be disadvantageous for smaller businesses. What I mean by this is that, I believe that, not only online shopping but technological advances in the modern world, would increase the already existing success gap between small businesses and colossal businesses, or even make companies which are large obsolete, reducing profits and increasing budgets.

To elaborate, with larger businesses being able to afford such high end technology for higher quality manufacturing and customer/worker satisfaction, the smaller businesses which could not afford such technology would be seen as a much worse business to work for or purchase from.

Which would change the public view of companies as a whole, making the large companies the best with quality and the smaller ones with cheaper production not as good, therefore not being able to sell its products. Which would in turn make it a lot harder to grow new businesses in the already existing market.

Culture (Ways of life, progressiveness, traditions, technology use/reliance)

In many ways, our culture has changed since the developments in HCI. For instance, in recent years people have become a lot more flexible in regards to their jobs. Basically, people can now work from home instead of travelling to an office. Especially with the new trend of tablets, laptops and mobile phones replacing physical notepads for example.

In other cases, the prevalence of technology has now led to us choosing easier, sometimes cheaper, methods of communication. Meaning that technology is beneficial not only to the user but to the environment too, theoretically speaking.

The reason that it might be beneficial to the environment is due to the fact it means we do not have to travel to go to work, therefore creating less carbon emissions. Hence why all of these factors have had an impact on the society we live in and are changing our culture. Basically making jobs become more efficient when more advanced technology is used and our culture is changing because of it, which some hope for the better.

Another aspect of culture would be the dependency on technology. Which means that people can access information for whatever they wish to know about, with websites like Wikipedia being a prime examples of this. Meaning that information is available to us at the click of a finger.

For example, Instant Messaging through applications, texting and social networks have reduced the need to meet people face-to-face. With emails also making the need for physical letters obsolete. Due to the letters being more expensive than sending emails and that emails can be sent from essentially any personal device such as a mobile phone, laptop or computer.

However, it also means that a culture with a lot of technology are then socialising less as they are usually working alone at home. Whether or not this is a positive or a negative change to our culture is subjective.

Another negative aspect is deskilling in our culture. In essence, deskilling means that new generations are lacking the skills to get jobs. As the technological world is developing, so to are the requirements to take jobs for IT and computing.

The older generations do not have the technological literacy to compete with younger generations, and younger generations are growing up to be lazy. This is bad for our culture, and bad for our image as a country.

Laziness itself can also be a large disadvantage to the culture. What I mean by this is that it’s often considered that, as generations get more and more technologically literate and dependent, people get lazier because things are easier.

To elaborate, with more automated systems and jobs being done by machines are resulting in humans doing less physical movement as a whole than in the past. With computers becoming self aware and intelligent, computer systems will take over more of our lives - thus making us as a society progressively more lazy.

Task 2

(U23D1) - Evaluate the impact of a potential future development in HCI

Task description - *It is important to think about the future, you are to* ***evaluate*** *in detail the potential negative and positive impacts that may occur in the future.*

A detailed report that evaluates at least one contemporary interaction model on society, culture and the economy

Future developments

Future developments in technology, especially those that revolve around Human Computer Interaction, is difficult to predict because it's hard to predict the future. Due to the fact that many have tried, but most have failed. The closest development which currently have many future predictions that I can personally think of in HCI is AI, which stands for Artificial Intelligence.

In the modern day, we've already developed robots for numerous purposes, yet we struggle to create humanoid AI that act like us. So far, we have created computer algorithms that learn as they go, and we've created driver-less cars.

With the technology we have in the modern day, we are still not quite at the Prometheus level of artificial humanoid intelligence, but we're getting closer. When it does reach that potential technological goal, AI will have a huge impact on the future of HCI.

As mentioned before, we've already taken steps towards creating useful AI, but we've still got quite a long way to go yet. However, with the technology we have today, AI will impact HCI by automating tasks that either we are incapable of doing, or are too dangerous/risky for humans to complete. Such tasks include bomb-defusing robots, and automated submersibles.

Also, AI will have an even bigger impact on our society, culture and economy. Especially with the introduction of recent wearable technology such as Google Glass, AI assistants (including Amazon's Echo, Apple's Siri, Microsoft's Cortana), wearable watches (which include pedometers, heartbeat sensors, and GPS trackers.

For example: the Apple Watch), and Virtual/Augmented Reality Headsets (Project Morpheus, Oculus Rift, HTC Vive), and mounted cameras (such as the GoPro 4) are all some of the big names in wearable technology.

These developments in technology will have their pros and cons, such as being lightweight with a decent functionality, for example. This would mean that jobs could be done more efficiently, but also that the users could lose weight and track their weight loss better meaning that society would be a generally happier, healthier place.

It would also theoretically reduce the chances of human slavery, since robots and AI do not need to get paid like other humans do. Essentially, instead of a salary they have maintenance costs. Which can be cheaper or more expensive, depending on the previous salary and the amount/quality of the robots/AI.

On the other hand, not all of the impacts of AI are positive on the culture today. For example, some people may argue that due to the ease and simplicity of interfaces in the workplace, key skills are being lost.

However, in the past everything had to be done by hand. Meaning that skilled men and women who used to work and make almost everything, would potentially lose their jobs, which are needed for day-to-day life. Therefore, overtime machines and computers will be, and in some areas have been, taking over the human workforce.

Now more than ever, the interaction between human and computers is less than in previous years, mainly found on industry machines and computers. This is because computers are becoming more self reliant and intelligent, so less people are needed to operate the same amount of machinery.

For instance, most of modern production are now already automated, so there will be people being laid off and in the end people will need to learn about system maintenance. Which will be considered a valuable skill needed if the worker is able to get any work in an automated future, besides some places where the work does not involve machinery.

Also, this is how I believe that AI will also impact the economy, since automation is becoming so prevalent. For instance, recently some suggestions were made by Amazon about drones and using them to deliver packages. Again, this has its ups and downs – it's a cheap, effective way of transporting some post to your door, but there is also the possibility of theft happening due to current AI's ability, or lack thereof, to predict.

However, even though AI can be used for very moral and positive charities for society, there are negative aspects that AI could bring to society, which are considered both immoral and controversial.

Such as AI used in the role of combat, through a military drone for example. For instance, a drone bring used for killing by a human controlling said drone, is considered controversial already because of the method of the killing.

However, if the drone has its own intelligence with an AI system and the fact that there are public fears with AI already, it would most likely cause a disturbance in public order. Especially, with pop culture demonising the idea of robots or AI that can turn against the human race, such as the popular movies “Terminator”, ”I, Robot”, “Blade Runner” and “2001: A Space Odyssey”, being the few movies that portray AI as an emotionless and dangerous entity to have.

Even though these popular movies being a work of fiction, some still believe that AI can posses a major threat to the human race. Which may be doubted by some and agreed by others, there is no doubt that a threat does exist with AI and the ability to eliminate at least a portion of the human population.

Even with extremely intellectual people working in science and technology, such as Bill Gates, Elon Musk and Stephen Hawking warning about the dangers of AI and how it can spell the end of the human race. Doubting those with a more optimistic view, paraphrasing: “They do not have a good enough understanding of how Artificial Intelligence works”.

In conclusion, it is still very uncertain whether AI will spell the end to the human race in the upcoming years, or whether it will make the population more unemployed and lazy due to an automated workforce. In my personal opinion, I believe that it is dangerous to mess with the power of Artificial Intelligence, especially for warfare.

To elaborate, I know that AI has the power to understand and process information much faster than the human brain, and that the evolution of AI to reach modern knowledge is miniscule compared to the evolution of mankind. Meaning that AI will reach a point where it takes less than a second to process something which would take a human a significant amount of time, such as a day, week or even a month.

It may sound ludacris now, but with technology constantly advancing at a consistent but rapid level, it would make the future a very dangerous gamble for the human race. Especially with the reliability of technology that our society, culture and economy has revolved around, even more to the implementation of AI into warfare.

However, I also believe that it is possible that AI would be somewhat less harmful towards humans. Even though they don’t technically have emotions to process empathy and kindness, it does also mean that they cant feel hatred or anger. It might be possible to process these emotions, but it wouldn’t be as genuine as a sentient being.

Overall, I believe that AI is a dangerous power to gamble with, especially with the ridiculous idea of implementing it into warfare. However, with the correct controllability and understanding the potential powers that can be abused, whether it is 10, 20 or 50 years, I believe that the future could be positive with AI for our society, culture and economy.